MILITARY SPECIFICATION

PUMPS, ROTARY, POWER DRIVEN, MISCELLANEOUS

This amendment forms a part of Military Specification MIL-P-19131 (SHIPS), 23 January 1956.

Page 1

1.2: Add :

"Type 11 - Screws with timing gears.
"Type X - Vane (slicing)
“Type XI - Slicing shoe.”

Pages 1, 2, 3 and 4

2.1: Delete reference to Military Specifications "MIL-T-15071", "MIL-N-17506", "NAVY DEPARTMENT General Specifications for Inspection of Material" and “General Specifications for Ships: Section S9-1 Welding and Allied Processes.”, and add the following:

“SPECIFICATIONS

“ FEDE RAL
"QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought

“MILITARY
"MIL-Q-9058 - Quality Program Requirements"
"MIL-M-15071 - Manuals, Equipment and Systems. ”

"STANDARDS

"MILITARY
"MIL-STD-278 - Fabrication, Welding and Inspection of Machinery Piping and Pressure Vessels for Ships of the United States Navy.

“PUBLICATIONS

“MILITARY
"250-648-8 - Inspection and Test of Silver-Brazed Piping Systems.
“0900-001-7000 - Fabrication and Inspection of Brazed Piping Systems.”

Page 5

Table I, column 5: Delete “33 - 100” and substitute “30 - 100”.

Page 6

3.3.1 item (e): Delete and substitute:

“ (e) Satisfactory operation when inclined as follows, unless otherwise specified in the contract or order:
(1) Up to 30 degrees from the normal horizontal position in any direction (permanently inclined).
(2) With the ship rolling up to 60 degrees from the vertical (time of complete roll cycle, 8 seconds) .

1/ CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE,

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(3) With the ship pitching 10 degrees up and down from the normal horizontal plane (time of complete pitch cycle, 4 seconds)."

Page 14

Add as paragraph 3.3.15.3:

"3.3.15.3 Horizontal flexibly coupled units with bedplates shall have couplings guards provided. Guards shall be designed to permit ready access to the coupling for lubrication and inspection."

Add as paragraph 3.3.16.7:

"3.3.16.7 When specified (see 6.1), pumps shall be provided with mechanical seals."

Page 15

3.3.19: Delete and substitute:

"3.3.19 Welding. Welding shall be in accordance with MIL-STD-278."

"3.3.19.1 Silver brazed joints shall be in accordance with NAVSHIPS 0900-001-7000. For joints 1/2 inch and above, only preinserted ring fittings shall be used."

Page 16

Table II, column 2 opposite "Shafts" add "17-4PH corrosion resisting steel."

Table II, column 3, line 7: Delete "MIL-N-17506" and substitute "QQ-N-286".

Page 17

3.4.3.1: Delete and substitute:

"3.4.3.1 Materials for class N-4 and O-3 pumps shall be the same as listed in table II except that commercial materials may be used in place of Government specification materials."

3.4.3.3. Delete and substitute:

"3.4.3.3 Materials of the vanes and sliding shoes of types IX, X and XI pumps shall be that moat suitable for classes N-2 and 0-2 services and shall be as proposed by the manufacturer and as approved by the command or agency concerned."

Page 18

3.5.3.1. Delete formula "$1.5=(\frac{\text{above formula hp.}}{4})$" and substitute:

$$"1.5-(\frac{\text{above formula hp.}}{4})"$$

Page 21

3.8.1: Delete and substitute:

"3.8.1 Manuals shall be in accordance with MIL-M-15071."

3.8.5, line 5: "MIL-T-15071" and substitute "MIL-M-15071".

Page 22

Table III: Delete and substitute;

"Table III - Onboard repair parts

Item	Name or part	Quantity per set
a	Ball or roller bearings	100 percent complete replacement for all installed units per ship
b	Oil seals	
c	Packing, if special, and packing spacers	
d	Coupling bushings, pins, springs and other wearing parts	
e	Mechanical seals	

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"Table III - Onboard repair parts (cent'c)

Item	Name of part	Quantity per set
f g	Sleeve, sliding surface bearings Cams, slides, slide pins, housings or liners, vanes (set) and similar wearing parts	One to four units per ship 100 percent replacement for one pump Over four units per ship 100 percent replacement for two pumps
h i	Timing gears, reduction gears and pinions Complete rotating element including bearings and complete coupling	100 percent replacement for one pump

3.9.2.1, line 1: Delete "item (f)" and substitute "item (g)".

Page 23

3.9.2.2, line 1: Delete "item (h)" and substitute "item (1)".

3.9.2.3, lines 1 and 2: Delete "items (a) and (g)" and substitute "items (a) and (h)", and in line 3, delete "item (h)" and substitute "item (1)".

Page 25

4.2.1: Delete and substikute:

"4.2.1 Qualification tests. Qualification tests on class N-2 and class O-2 pumps shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. These tests shall Consist of the test specified in 4.2.8."

4.2.2: Delete and substitute:

"4.2.2 In addition to the tests at the laboratory class N-2 and O-2 pumps may be required to pass the following shipboard service test:

(a) Two or more pumps of the same general design and general operating characteristics shall have given satisfactory service onboard Naval ships for a period of at least two years and shall still be in service on such ships at the time qualification approval is granted."

Page 27

4.3.2, lines 1 and 2: Delete 'as selected by the Government Inspector".

Page 28

4.3.3.1: Delete class "O-3" and substitute class "O-2; line 2, delete "tested on order" and substitute "tested in order".

4.3.3.2, line 1: Delete class "11-2" and substitute class "N-2".

4.3.4.1, line 2: Delete "by the representative of the bureau or agency concerned."

Page 30

4.3.5, first sentence: Delete and substitute "Sufficient data shall be taken during the tests of 4.3.1, 4.3.2 and 4.3.3 to prepare pump characteristic curves as specified below."

Page 33

4.5: Delete and substitute:

"4.5 Inspection and test of *silver* brazed joints in systems designated as hazardous in manual NAVSHIPS 250-648-8 shall be in accordance with the requirements of that manual."

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Add as paragraphs 4.6 through 4.6.1.3:

"4.6 QUALITY ASSURANCE PROVISIONS

"4.6.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of tile inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and *services* conform to prescribed requirements.

"4.6.1.1 Inspection system. The inspection system which the supplier is required to maintain, as provided in the inspection clause of the contract or order, shall be in accordance with MIL-I-45208.

"4.6.1.2 All instrument used in performance acceptance tests shall be calibrated before and after tests.

"'4.6.1.3 No instrument containing mercury shall be used in tne manufacture or testing of any equipment destined for installation on a nuclear powered ship. This point shall be covered in the manufacturer's quality control manual. Test reports shall contain a certification. signed by a responsible official of the manufacturer that no mercury containing instruments have been used in the manufacture or testing of the equipment."

Page 35

6.1, item (h)(9): Delete and substitute.

"(h)(9) Whether mechanical shaft seals are required (see 3.3.16.7)."

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